AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM340
Natural Language Processing

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

Multiple Choice Questions **(15 marks)**

Question 1, (15 marks, 1.5 mark for each question)
Answer the following MCQ's by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet.

1. The Turing Test was designed to empirically test if a machine could do what?
    a) Perform calculations faster than a human.
    b) Think and use language indistinguishably from a human.
    c) Store a large amount of text data.
    d) Translate between two human languages flawlessly.

2. What is the primary purpose of using a **backtrace** in the Minimum Edit Distance algorithm?
    a) To find the actual sequence of edits (insertions, deletions, substitutions) that gives the minimum cost.
    b) To calculate the final distance score more quickly.
    c) To initialize the distance matrix before the calculation begins.
    d) To apply different weights to different types of edits.

3. What is the primary motivation for adding a regularization term in a logistic regression model?
    a) To increase the model's performance on the training data.
    b) To penalize large weights and reduce overfitting.
    c) To speed up the training process using stochastic gradient descent.
    d) To enable the model to handle more than two classes.

4. What is the primary difference between **lemmatization** and **stemming**?
    a) Stemming is done by morphological parsing, while lemmatization is done by removing affixes in a basic way.
    b) Only lemmatization can handle uppercase letters.
    c) Stemming crudely chops off word endings, while lemmatization reduces words to their actual dictionary form.
    d) Stemming uses a dictionary, whereas lemmatization uses rule-based heuristics.

5. The **Markov Assumption** simplifies N-gram models by stating that the probability of a word depends on what?
    a) Only a fixed number of preceding words.
    b) All previous words in the sentence.
    c) The total number of words in the vocabulary.
    d) The emotional sentiment of the sentence.

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6. If a classifier is trained on a dataset where 99% of documents belong to Class A, why is **accuracy** a poor evaluation metric?
    a) Accuracy cannot be calculated when there are more than two classes.
    b) A high accuracy can be achieved by a useless classifier that simply always predicts the majority class.
    c) Precision and recall are mathematically equivalent to accuracy in this scenario.
    d) The F1-score will always be zero, making accuracy the only available metric.

7. What problem in language modeling is **Laplace (add-one) smoothing** designed to solve?
    a) The issue of floating-point underflow from multiplying many small probabilities.
    b) The problem of assigning a zero probability to events not seen in the training data.
    c) The difficulty of handling very large vocabularies.
    d) The inability of N-gram models to capture long-distance dependencies.

8. In regular expressions, which character makes the preceding element optional (matching zero or one time)?
    a) *
    b) +
    c) ?
    d) .

9. Which of the following is considered a **discriminative** classifier?
    a) Naive Bayes
    b) Logistic Regression
    c) Language Models
    d) Hidden Markov Models

10. In the **Byte-Pair Encoding (BPE)** algorithm, how are new tokens created?
    a) By splitting the least frequent words into characters.
    b) By randomly selecting pairs of characters from the corpus.
    c) By merging the most frequently adjacent pair of symbols in the training corpus.
    d) By adding all out-of-vocabulary words from a test set.

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PART 2

Short Essay Questions **(15 marks)**

Question 2, (15 marks)

Answer the following questions:

a) What is the goal of the 'Data Pre-processing' stage in a generic NLP pipeline? Name two common tasks performed during this stage. **[5 marks]**

b) In text processing, what is the difference between a "**word token**" and a "**word type**"? Provide a simple example to illustrate. **[5 marks]**

c) What is the "**zero probability**" problem in N-gram language modeling? Explain why this is a critical issue when evaluating a model using perplexity? **[5 marks]**

PART 3

Short problems/questions **(30 marks)**

Question 3, (10 marks)

A spell-checker needs to determine if the misspelled word **'slep'** is more likely intended to be **'sleep'** or **'step'**.

Using the **Weighted Minimum Edit Distance** algorithm, determine which of the two correct words is closer to **'slep'**. Assume the following costs, which are based on common typos:

- All **insertions** and **deletions** cost 1
- All **substitutions** cost 2, **except** for the substitution of 'l' for 't' (a common error), which costs only 0.5

You must:

a) Write the weighted distance matrix for transforming **'slep'** into **'sleep'**. **[3 marks]**

b) Write the weighted distance matrix for transforming **'slep'** into **'step'**. **[3 marks]**

c) State the final minimum edit distance for both pairs. **[2 marks]**

d) Conclude which word is closer to **'slep'**. **[2 marks]**

Question 4, (10 marks)

Given the following small training corpus:

<s> the sun is hot </s>
<s> the moon is cold </s>
<s> a hot sun </s>

And the test sentence **S**: <s> the sun is cold </s>

Answer the following (**Show the formula and calculations**):

a) Calculate the bigram probabilities for each word pair in the test sentence **S** using **Maximum Likelihood Estimate (MLE)**. **[4 marks]**

b) Calculate the **probability** of the test sentence **S**. **[2 marks]**

c) Calculate the **perplexity** of the test sentence **S**. **[2 marks]**

d) Suppose that you want to generate new text using the provided corpus. If you have just sampled '**hot**', what are the possible next tokens you could sample, and what is the probability of each one? **[2 marks]**

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Question 5, (10 marks)
Assume a **Naive Bayes classifier** for **spam detection** with the following known likelihoods:

Word	Spam (+)	Not Spam (-)
winner	0.30	0.01
money	0.20	0.05
meeting	0.02	0.25
report	0.03	0.20

From the training data, you know the following about the distribution of class:

- Total documents: 200
- Number of Spam (+) documents: 50
- Number of Not Spam (-) documents: 150

And given the document d = "**money report**",

Answer the following (show your calculations for each step):

a) Calculate the prior probability for each class? **[3 marks]**
b) Calculate the posterior probability score for each class? **[5 marks]**
c) What class will Naive Bayes assign to the document d? **[2 marks]**

End of questions

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